

Maneuver Planner for Automated Vehicles on Urban Scenarios

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Abstract—This work presents a maneuver planner for automated vehicles, seamlessly integrated with a motion planner, that is able to make tactical decisions to safely navigate in urban environments. The proposed maneuver planner is able to select the most suitable target lane for the ego-vehicle and to manage different regulatory signals like right of way, must stop or pedestrian crossings. The integration with the motion planner allows to select the current maneuver after exploring a reachable set of trajectories, which brings flexibility to the system in complex and dynamic driving scenarios. The architecture was validated on a real vehicle in two urban scenarios with other traffic agents: an intersection with a pedestrian crossing and an overtaking scenario on a two-way road.

I. INTRODUCTION

Autonomous Driving Systems (ADS) must constantly make tactical decisions in order to be efficient, specially in urban scenarios. These decisions may include: changing to a more convenient lane given a global route, stopping on a yielding line on an intersection, waiting for a pedestrian to cross the road or taking the next exit ramp on a highway. To achieve these kind of complex tasks, a two-level hierarchical architecture formed by a maneuver planner and a motion planner has been widely deployed [1]. This modular architecture has different advantages, like higher flexibility when testing different algorithms, easier maintenance due to the independence of the modules, and better scalability [2]. The tactical or maneuver planning, according to [3], may be classified into three categories: (i) rule-based tactics, which consider speed limits, working zones, stop signs or intersection precedence handling; (ii) route-based tactics, which consist of choosing the lane in order to meet global routing requirements; and (iii) maneuver-based tactics, which decide when and how to yield, follow or overtake other agents.

In this work it is proposed a maneuver planner that comprises both route-based tactics and maneuver-based tactics. The maneuver planner is integrated with a motion planning system, which explores a set of reachable trajectories and the resulting trajectory set contributes on the decision making process. This architecture (depicted on Fig. 1) brings flexibility to the ADS and deals, as it is not limited by pre-designed maneuver-based tactical rules, and is therefore able to efficiently handle unforeseen situations [3].

Most of the Decision Makers (DM) for autonomous driving described in the literature focus on one kind of

scenario, i.e. city intersections, lane-changing on a highway or overtaking maneuvers. In [4], the authors propose a rule-based DM to solve intersections by finding conflicting points between the Ego-Vehicle (EV) path and the Other Vehicles (OV) present in the scene, to define a drive/stop policy. In [5] a motion planner module generates speed profiles based on human drivers data, while the DM selects the maneuver that minimizes the risk of collision. On [6] the authors implemented a Finite State Machine (FSM) which varies the speed profile of the EV according to its current state in order to solve T-intersections. Finally, in [7], the proposed DM also manages T-intersections, only this time using a discrete event handler with limited visibility and no communications with other OVs.

Alternatively, some DM handle only lane-changing maneuvers. For example, in [8] it is presented a DM based on an FSM to perform lane changes by taking into account the global route as well as the current driving environment. [9] propose to perform overtaking maneuvers on freeways using a two stage architecture, consisting on a lane-change decision trigger, based on a Radial Basis Function (RBF) neural network, and a hierarchical state machine to complete the maneuver. This kind of systems show good results in the environments for which they were designed, but may struggle to solve different urban scenarios.

There are some works that implemented a DM using neural networks [10] [11] [12] [13], but this kind of systems rely strongly on the data quantity and quality and have a high training cost. In addition to that, their black box nature

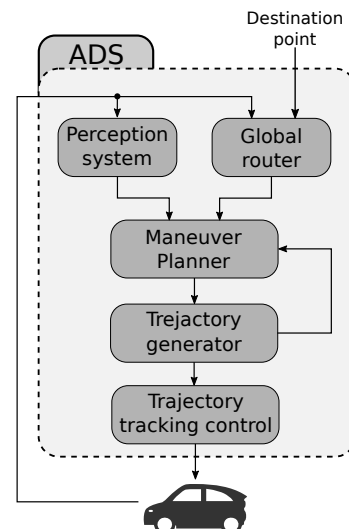


Fig. 1. Complete ADS architecture.

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makes their functional safety compatibility [4] and semantic interpretation [14] challenging.

The maneuver planner proposed in this work selects the target lane and establishes a number of restrictions according to the regulatory signals present in the scene. A regulatory signal establishes a set of rules that must be followed by the traffic agents in the road; some examples are must-stop signals, pedestrian crossings or right-of-way signals on intersections. Then, the motion planner generates a set of reachable trajectories Γ , which is used by the maneuver planner to evaluate the feasibility of the proposed maneuver, and to change it if necessary. Also, different kind of maneuvers like merging or yielding into a roundabout may be executed depending on the reachable trajectories. A detailed description of the trajectory generator is provided in section II-D. The majority of decisions on city and highway driving scenarios can be covered using this approach, since (i) most of them can be decomposed on basic driving modes like lane changing or stopping on yielding lines [2]; (ii) the integration with the motion planner provides a necessary feedback in dynamic environments to determine whether the maneuver is feasible to the EV or not.

The main contributions of this work are:

- A hybrid decision making architecture that allows to perform tactical decisions using a seamlessly integrated maneuver planner and trajectory generator, which prevents conservative behaviors.
- The proposed architecture is not designed to handle only a specific type of scenario but it can handle multiple types of urban scenarios by combining lane changes and regulatory signal handling on a tactical level.
- The system was tested on a real vehicle on different urban scenarios with other traffic agents.

The outline of the paper is as follows: section II describes the maneuver planner in detail, section III shows the results in real-life scenarios and section IV presents some concluding remarks.

II. MANEUVER PLANNER

Fig. 2 depicts the architecture of the maneuver planner and its integration with the trajectory generator. This system uses the perception information Ω and the global route φ to calculate a set of way-points ω , a traffic-regulation state S and the identification number id of a potential regulatory signal. The trajectory generator computes the trajectory set Γ , and provides the number of trajectory candidates $\tau \in \Gamma$ considered valid for the current scene (N_{valid}). A trajectory τ is considered valid if it does not leave the road at any moment and it does not lead the EV to a collision with other traffic agents. If $N_{valid} < \eta$, (being η the minimum number of valid candidates to consider a maneuver as feasible) during a lane change maneuver, the maneuver planner cancels the maneuver, updates the target corridor and re-calculates ω , in order to guarantee a feasible maneuver.

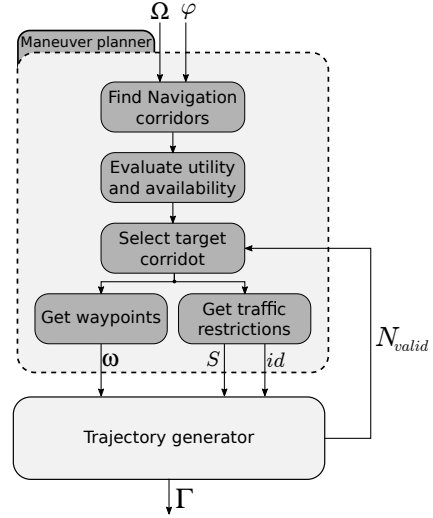


Fig. 2. Maneuver planner scheme and its interaction with the trajectory generator.

A. Navigation corridors

The first task performed by the maneuver planner is to identify the set of available navigation corridors ζ . Each navigation corridor ζ^i is a reachable lane for the EV, given its location on the lanelet2 digital map [15]. This type of map provides two layers of information: physical and topological. The physical layer contains geographic information like borderlines, centerlines or location of elements, while the topological one contains the relation between road elements in a graph network. ζ^i is computed by finding the lanelet(s) where the ego-vehicle is located. Next, a graph search is performed in the topological layer of the map to obtain a lanelet sequence. Each corridor is described with three different types of information: routing, topological and traffic. Regarding the routing information, ζ^i stores the id of the lanelets, the distance to the next exit, and the compliance with the global route (a boolean value that indicates if the corridor is contained in the global route or not). Concerning the topological information, ζ^i contains the centreline and the borderlines of the lane. The traffic information stored in ζ^i is composed of the speed and position of the closest lag and lead vehicles.

B. Selection of the target corridor

The target corridor ζ^t is selected using a lane-change model proposed in [16]. It proposes a utility function that evaluates mandatory and discretionary metrics for each corridor, such as average speed, next-exit distance or relations with respect to the OVs. The utility value U is calculated for each corridor ζ^i using this expression:

$$\begin{aligned}
U_{\zeta}^i = & \beta_{\zeta}^i - C^1 D_{\zeta}^i + C^2 S_{\zeta}^i + C^3 \Delta X_{\zeta}^{lead,i} + C^4 \Delta S_{\zeta}^{lead,i} \\
& - C^5 \delta_{\zeta}^{tg} \delta_{\zeta}^{cl,i} + C^6 \delta_{\zeta}^{cl,i} - C^7 \Delta L C_{\zeta}^i \\
& + \beta_{\zeta}^{path,i} [d^{exit}]^{-C^8} - C^9 \delta^{ne} \Delta Exit_{\zeta}^i \\
& - \alpha_{\zeta}^i v_n - C^{op} \delta_{\zeta}^{op,i} \quad (1)
\end{aligned}$$

where the variables definition is as follows: β_{ζ}^i is a constant depending on the right-to-left location of the corridor; D_{ζ}^i and S_{ζ}^i are the density of vehicles and the average speed, respectively; $\Delta X_{\zeta}^{lead,i}$ and $\Delta S_{\zeta}^{lead,i}$ are the separation and relative speed with respect to the lead vehicle, respectively; $\delta_{\zeta}^{cl,i}$ is a boolean variable that is non-null if ζ^i is the current lane of the EV, while $\delta_{\zeta}^{tailgate}$ is activated if the EV has a tailgate vehicle; $\Delta L C_{\zeta}^i$ is the number of lane changes required to get to ζ^i ; $\beta_{\zeta}^{path,i}$ is the plan impact coefficient, which increments with the number of lane changes required to reach the next exit on the global route, while d^{exit} is the distance to that exit. δ^{ne} is a boolean variable with value 1 if the driver intends to take the next exit and 0 otherwise; $\Delta Exit_{\zeta}^i$ is the number of lane changes required to go from ζ^i to the exit lane; The term $\delta_{\zeta}^{op,i}$ indicates if ζ^i is a two way lane and may be used by the EV to perform an overtaking maneuver. Notice that this term, and its coefficient C^{op} were not included on the original model and were included in this work to allow the ADS to perform overtaking maneuvers on two-way roads. The weighting coefficients C^i , $i \in 1...9$ of each variable is defined in [16].

Once the utility of ζ is computed, the probability of being selected as ζ^t is computed for each ζ^i using the multinomial logit model:

$$P(\zeta^t = \zeta^i) = \frac{\exp(U_{\zeta}^i)}{\sum_{j \in |\zeta|} \exp(U_{\zeta}^j)} \quad (2)$$

The maneuver planner sets as ζ^t the corridor with the largest probability that meets two requirements: (i) it has a valid gap acceptance and (ii) all its lanelets are included in the global route. The gap acceptance of an adjacent corridor depends on the lead gap ΔX^{lead} (space between the lead vehicle and the EV) and the lag gap ΔX^{lag} (space between the EV and the lag vehicle). Both values have to be greater than a corresponding critical gap to be valid. The models to obtain the gap acceptance of a corridor are detailed in [16].

C. Regulatory signal handling

The maneuver planner is able to handle three types of regulatory signals: Must Stop (MS), Right of Way (RW) and Pedestrian Crossing (PC). The MS signals are formed by a yielding lanelet and a Yielding Line (YL). In order to comply with this kind of signal, the EV must stop completely before reaching the YL. The RW signal is formed by a YL, one or more yielding lanelets and one or more priority lanelets. For this type of signal, the vehicle that gets into the intersection from a yielding lanelet has to stop on the YL if there is another vehicle on the priority lanelet, unless it is able merge

before that vehicle in a safely manner. This type of signals are commonly used on crossroads or roundabouts. The PC signal is formed by a crossing zone, one or more YL and one or more yielding lanelet. In this case, if there is a pedestrian on the crossing area, the vehicle must stop before the YL and it must remain stopped as long as there is a pedestrian on the road. A regulatory signal $\psi^r \in \Psi \forall r = \{0, ..., R-1\}$, with Ψ being is the finite set of R regulatory elements on the global route, contains different information like the unique id, the associated yielding lines, the lanelets of interest or the crossing area polygon; all of this data is embedded on the digital map. Fig. 3 depicts a driving scenario with two regulatory signals ψ^1 and ψ^2 . On the one hand, the signal ψ^1 is a RW signal with one yielding lanelet, which is highlighted in red, two priority lanelets, which are plotted in green and a yielding line plotted in yellow. On the other hand, ψ^2 is a PC signal with two yielding lanelets, two yielding lines and a crossing zone for pedestrians. The crossing zone includes also the sidewalk and not only the road, so that the EV is forced to stop if there is a pedestrian waiting to cross.

A finite state machine with four possible states $S = \{Go, Yield, Try, Aware\}$ is used to comply with the aforementioned regulatory signal behaviors. The current state S_k is updated taking into account if there is a regulatory signal ψ^t on ζ^t and also the gap distances with the OV's of the traffic scene. Fig. 4 depicts the state diagram of the FSM proposed to handle regulatory signals. The initial state is $S_0 = Go$, and the future states are obtained according to the state-change conditions defined in Table I. When the state S_k is updated from any other state to $S_{k+1} = Go$, the regulatory signal ψ^t is omitted and it is not taken into account again until it is completely crossed. This omitted signal will be referred to as ψ^o . Each signal ψ^r is described using three variables: tp_{ψ}^r defines the type of signal, id_{ψ}^r is the unique id on the digital map, and g_{ψ}^t is the distance from the EV to the yielding line. In case $tp_{\psi}^r = PC$ two extra variables are used: tzf_{ψ}^t is the time the crossing zone has been free of pedestrians, being set to 0 when a pedestrian is detected on the crossing zone; and trf_{ψ}^t , which is the time the road has been free while the EV is stopped on a PC.

d_o is a design parameter that defines the minimum separation between the EV and ψ^t before it can be omitted. d_c is another design parameter that establishes the minimum acceptable

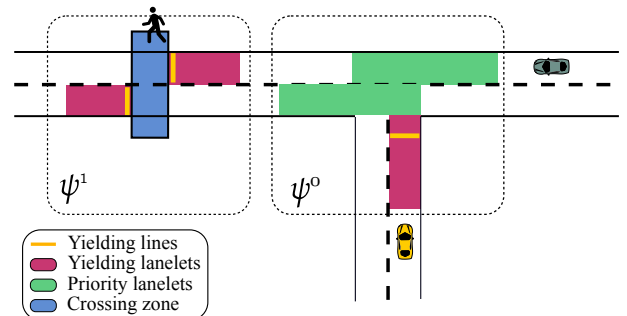


Fig. 3. Regulatory signals components on an urban scenario.

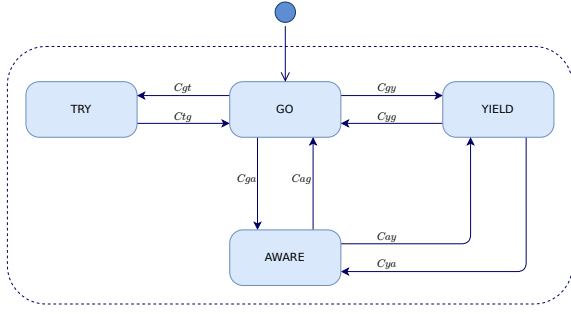


Fig. 4. Finite state machine for regulatory signal handling.

TABLE I

STATE TRANSITION CONDITIONS FOR THE REGULATORY SIGNAL FSM.

Name	Definition
c_{gt}	$tp_{\psi}^t = RW \wedge id_{\psi}^t \neq id_{\psi_o}$
c_{tg}	$g_{\psi}^t < d_o \wedge \Delta X_{\zeta}^{lead,t} > d_c$
c_{ga}	$tp_{\psi}^t = PC \wedge id_{\psi}^t \neq id_{\psi_o}$
c_{ag}	$g_{\psi}^t \leq 0$
c_{ay}	$tzf_{\psi}^t = 0$
c_{ya}	$tp_{\psi}^t = PC \wedge tzf_{\psi}^t \geq t_{o1} \wedge g_{\psi}^t > d_o$
c_{gy}	$tp_{\psi}^t = MS \wedge id_{\psi}^t \neq id_{\psi_o}$
c_{yg}	$\left(tp_{\psi}^t = MS \wedge g_{\psi}^t < d_o \wedge v_{ego} = 0 \wedge \Delta X_{\zeta}^t > d_c \right) \vee$ $\left(tp_{\psi}^t = PC \wedge g_{\psi}^t < d_o \wedge trf_{\psi}^t \geq t_{o2} \right)$

gap with respect to the leader vehicle to cross a yield line. t_{o1} and t_{o2} are different times thresholds which are used to update S_k when a PC signal is found.

When the current traffic-regulation state is $S_k = Go$, if $tp_{\psi}^t = RW$, the new state is $S_{k+1} = try$. In this state, the trajectory generator tries to find a feasible speed profile that allows the EV to merge before a vehicle with higher priority, or to stop on the YL if it is not possible. If $S_k = Go$ and $tp_{\psi}^t = MS$, then $S_{k+1} = yield$, leading the speed profile of each path candidate to always stop before reaching the YL. Lastly, if $S_k = Go$ and $tp_{\psi}^t = PC$ then $S_{k+1} = aware$; in this state, the trajectory generator reduces the speed before crossing the signal. If $s_k = aware$ and there is a pedestrian detected on the crossing zone, then $S_{k+1} = yield$, and the EV will stop before the YL.

D. Integration with the trajectory generation

The trajectory generator creates N trajectory candidates and selects the best of them using a merit function (see [17] and [18] for more details). The paths are created using quintic Bézier curves starting from the current EV trajectory and ending on different way-points ω^j located on the centreline of ζ^t . The speed profile for the candidates is generated taking into account a set of comfort constraints, the position of the other traffic agents and the potential traffic regulations that may be imposed by ψ^t . A candidate is considered valid if it is collision free and it does not occupy any space outside the road.

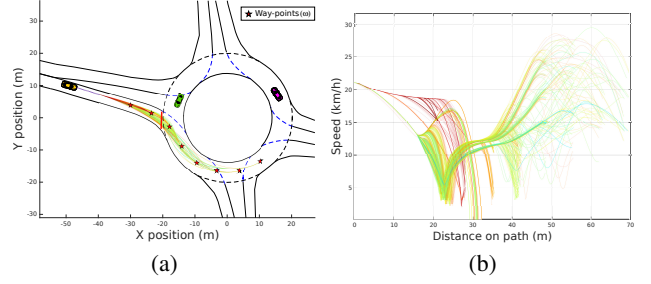


Fig. 5. Merging into a roundabout. (a) Set of way-points ω and path candidates. (b) Speed profiles for the candidates.

Fig. 5a shows the way-points ω and the path candidates generated for a roundabout scenario where $S_k = try$ and $g_{\psi}^i = 34m$. Fig. 5b depicts the speed profiles generated for each valid candidate. In this case, some speed profiles stop before reaching the yielding line and other speed profiles merge into the roundabout between the green and the magenta vehicles without stopping. The final maneuver is therefore chosen by the trajectory generator, following the candidate selection described in [18].

The number of valid candidates created by the trajectory generator during a lane change maneuver is used as a feedback indicator by the maneuver planner, and, in case there is not at least η valid candidates, ζ^t is updated to cancel the lane change and to keep the current lane.

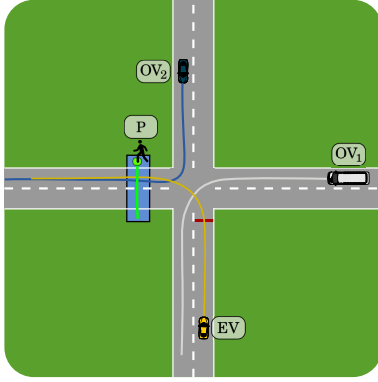
III. EXPERIMENTAL RESULTS

The complete ADS was implemented on a real vehicle in order to test the maneuver planner architecture on two scenarios with real traffic agents. The instrumented vehicle is a fully automated Citroën DS3 with different on-board sensors (see a detailed explanation of the experimental platform in [19]).

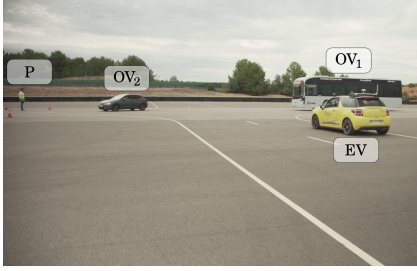
Two scenarios were selected to test the system. In the first one, the EV and two other vehicles (OV_1, OV_2) are approaching an intersection from different roads. The EV has to perform a left turn and there is a PC signal after the intersection, where a pedestrian (P) will cross when the EV approaches the signal. The setup of this scenario is depicted in Fig. 6a. Fig. 6b shows a capture of the real experiment on the testing track.

In the second scenario the EV has to overtake another vehicle (OV_1), while on the opposite lane there is an oncoming vehicle (OV_2); hence, the EV has to wait until the opposite lane is free before performing the overtaking maneuver. Fig. 7a shows the setup of this scenario and Fig. 7b shows the status of the vehicles at a specific moment during the experiment. Note that to increase the visibility of the EV and therefore be able to detect OV_2 well in advance, a collective perception architecture was implemented between the EV and OV_1 [20].

The configuration parameters used in the experiments for the state transition conditions given in table I are shown in Table II.

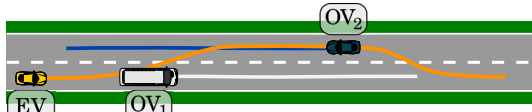


(a)



(b)

Fig. 6. Intersection experimental scenario. (a) Experiment setup. (b) Traffic status during the real experiment.



(a)



(b)

Fig. 7. Overtaking scenario. (a) Experiment setup. (b) Traffic status during the real experiment.

TABLE II
CONFIGURATION PARAMETERS FOR TESTING.

Parameter	Value	Units
d_o	9	m
d_c	7	m
t_{o1}	3	s
t_{o2}	1	s

By using these parameters, the EV will consider to discard a regulatory signal when it is closer than 9m from the corresponding yielding line. The distance with respect to a leader vehicle has to be greater than 7m to consider discarding a stop or a RW signal. Also, the EV will start to drive over a PC signal if the road has been free for over 1 second.

The complete video demonstrator for the intersection scenario can be accessed in www.youtube.com/watch?v=iOn0VP7fZ3A, while the one for the overtaking scenario is accessible in www.youtube.com/watch?v=0F4ov41tFE4

A. Intersection scenario

In this scenario the EV has to drive over two different regulatory signals before reaching the destination. The first of them (ψ^0) is a RW signal, where the EV is approaching from the yielding lane so that OV_1 has higher priority. The second signal (ψ^1) is a PC signal with one person on the crossing area. Fig. 6a shows the yielding line of ψ^0 as a red line and the crossing zone of ψ^1 as a blue rectangle.

The evolution of S_k and the EV speed during the experiment are plotted in Fig. 8. To provide a higher context to the figure, three events were highlighted using vertical dotted lines: the crossing of the yielding line of ψ^0 , the moment where the road over the PC signal has been free for over t_1 seconds after the EV has stopped, and the crossing of the yielding line of ψ^1 .

The EV detects the RW signal ψ^0 immediately after it starts (see Fig. 9a), leading to a state change from $S_k = Go$ to $S_{k+1} = try$. While in the latter state, the trajectory generator did not find a feasible trajectory to merge before OV_1 reached the intersection, so the EV stopped on the yield line during the interval $t = [6.8s, 13s]$ (Fig. 9b). The condition c_{tg} was triggered at $t = 13s$ because OV_1 did not intersect the EV trajectory any longer, hence $\Delta X^{lead} \rightarrow \infty$. At that moment $S_k = Go$ and $id_{\psi^0} = id_{\psi^1}$ and, by doing this, this signal is omitted until its YL is completely crossed. Since ψ^0 is no longer taken into account, the next signal ψ^1 is detected (the PC signal), the state changes to $S_k = Aware$ and then to $S_k = yield$ because there is a pedestrian located in the crossing area. The EV starts to move and crosses the first yielding line of ψ^0 at $t = 17.1s$ (Fig. 9c) and yields before the yielding line of ψ^1 . The EV maintains the yield

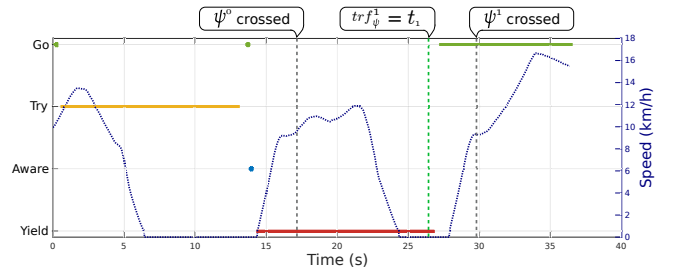


Fig. 8. Traffic-regulation state S and speed profile all along the intersection experiment.

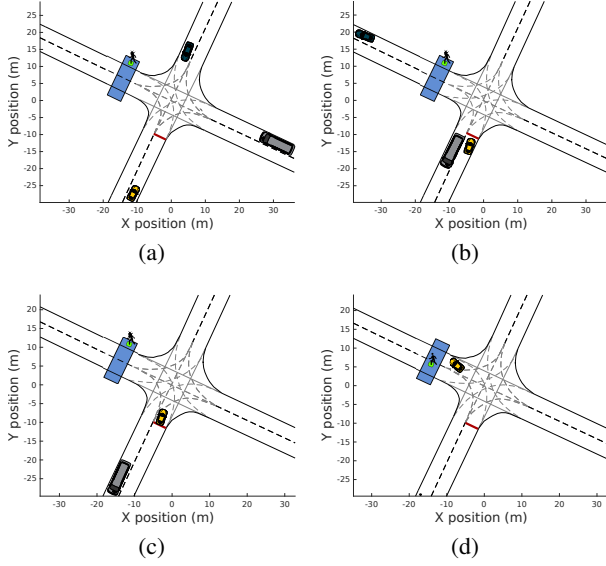


Fig. 9. Time evolution of the intersection traffic scene. (a) traffic status at $t = 1s$. (b) traffic status at $t = 13s$ (c) traffic status at $t = 17.1s$ (d) traffic status at $t = 27.1s$.

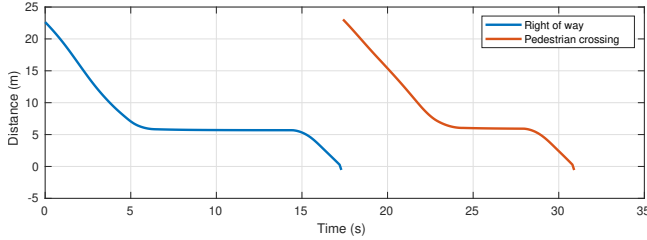


Fig. 10. Distance to the yielding lines of ψ^0 and ψ^1 during the intersection scenario.

decision until $t = 27.1s$, when the pedestrian exits the lane, so that $trf_{\psi}^1 \geq t_1$, triggering the condition c_{yg} (Fig. 9d). From that moment, and until the end of the experiment, the signal status is $S_k = Go$.

The scenario was correctly handled and the behavior was appropriate in both signals, keeping at all times a safe driving profile. Fig. 10 shows the distance to the yielding lines of ψ^0 and ψ^1 . It can be seen that the EV stopped at a distance of 5m in both cases.

B. Overtaking scenario

In this scenario, the EV has to safely overtake another vehicle. There are two possible corridors for the EV in this case. The first one is the right corridor (ζ^0), which is also the initial corridor of the EV. The second one is the left corridor (ζ^1), which is an opposite-way lane. Fig. 11 shows the probability assigned to each corridor during the trip, using a colour code to represent their status, as follows: blue indicates when $\zeta^i = \zeta^t$; green means ζ^i is available, but $\zeta^i \neq \zeta^t$; and grey is referred to the situation where the gaps of the corridor are not valid at the moment.

At the beginning of the experiment, the right lane is selected as the target corridor since $P(\zeta^0 = \zeta^t) = 0.66$. At this moment, the gap with respect to OV_2 is valid, but

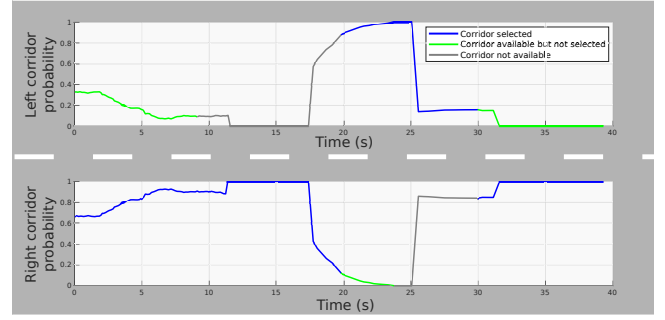


Fig. 11. Probability of $P(\zeta^i = \zeta^t)$ for the right and left corridors during the overtaking scenario.

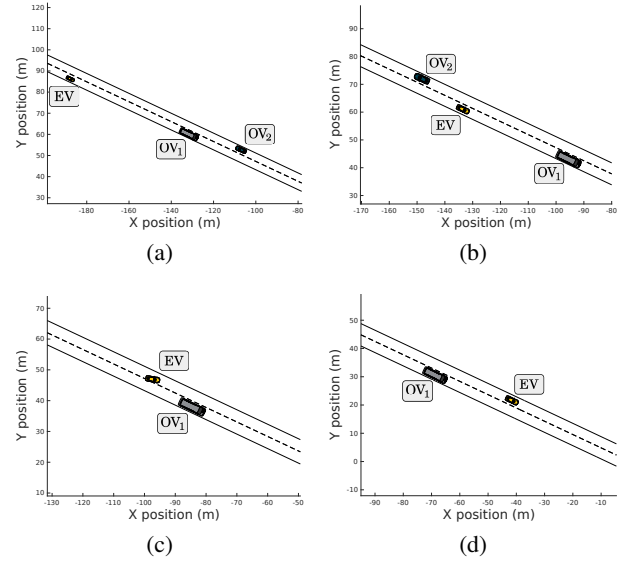


Fig. 12. Time evolution of the intersection traffic scene. (a) traffic status at $t = 8s$. (b) traffic status at $t = 19.5s$ (c) traffic status at $t = 23.7s$ (d) traffic status at $t = 30s$.

the value of $P(\zeta^1 = \zeta^t)$ is low. At $t = 8s$ (see Fig. 12a), the gap of ζ^1 is not considered valid anymore and this corridor becomes unreachable until $t = 19.8s$ (Fig. 12b), when the target corridor changes also, as $P(\zeta^1 = \zeta^t) > P(\zeta^0 = \zeta^t)$, triggering a lane-change maneuver. During the overtaking maneuver, the gap of ζ^0 is not valid due to the proximity between the EV and OV_1 (the status of the vehicles at this moment is depicted on Fig. 12c). When the EV overpasses OV_1 at $t = 25.5s$, the probability of the target corridor changes notably, and $P(\zeta^1 = \zeta^t) < P(\zeta^0 = \zeta^t)$, but the gap acceptance model does not allow to select this corridor until $t = 30s$ (Fig 12d), where the $\Delta X^{lag} = 31.5m$. At this moment, the EV starts the lane changing maneuver to the original lane, in order to complete the overtaking maneuver.

The gaps between the EV and the vehicles OV_1, OV_2 during the experiment are plotted in Fig. 13. Every positive gap means that the OV_i is ahead the EV. For this experiment, OV_1 started 45m ahead the EV, and OV_2 started at 120m. Four time instants are highlighted on the figure using vertical dotted lines, which match the four frames plotted on Fig. 12.

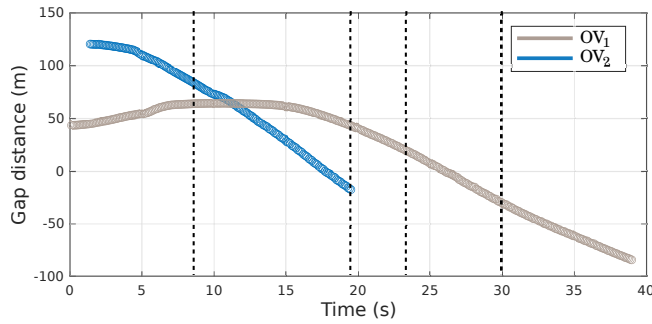


Fig. 13. Gaps with respect to OV_1 and OV_2 during the overtaking scenario.

IV. CONCLUDING REMARKS

A maneuver planner for automated vehicles was designed and implemented within a hierarchical architecture to take tactical decisions about lane selection and regulatory signals handling. The lane-selection algorithm uses a utility model which considers both the current traffic status and the global route, while it ensures safety by applying a gap-acceptance model. The regulatory signal handling is performed using an FSM that can manage right of way, must stop and pedestrian crossing signals. The maneuver planner establishes a set of restrictions which are taken into account by a motion planner to generate a reachable trajectory set, and finally, a valid trajectory for the EV. The trajectory set is involved in the decision making process, providing flexibility and robustness to the system in complex driving scenarios. The complete ADS was implemented on a real vehicle and it was successfully validated on two real scenarios: an intersection with two other vehicles and a pedestrian; and a two way road where the EV had to perform an overtaking maneuver.

Future work will be oriented towards the incorporation of other regulatory signals as traffic lights or reserved lanes. Also, the maneuver planner may include the integrity of the sensory data to propose different type of maneuvers accordingly i.e. if the quality of the lidar information is not reliable enough, lane changes are avoided.

ACKNOWLEDGMENT

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